

Deeraj Nagothu, Ph.D.

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Professional Summary

Ph.D. Research Scientist specializing in production AI systems: LLM-powered agentic AI, transformer architectures, foundation model deployment, and AI safety. 7+ years building scalable ML solutions from research to production: developed neuro-symbolic agents integrating LLMs/VLMs with symbolic reasoning, deployed real-time inference pipelines (Airflow, Kubernetes), and engineered distributed AI frameworks for cloud/edge environments. Published 30+ papers while shipping production-ready code including open-source libraries. Applying deep expertise to advance LLM engineering and agentic AI capabilities.

Technical Focus Areas

- ◇ LLM Development & Optimization: model training, transformer architectures, inference optimization, prompt engineering, and Agentic AI systems. Model evaluation, adversarial robustness & deepfake detection.
- ◇ Computer Network security, Blockchain and Virtualization.

Work Experience

- 2023 – Present ◇ **Research Scientist.** Intelligent Fusion Technology Inc. (IFT), Germantown, MD, USA.
- Architected and deployed a transformer-based computer vision agentic pipeline integrating Stable Diffusion XL with Vision-Language Models (VLMs) for automated prompt engineering, improving YOLOv11 object detection recall by 33% on low-visibility targets.
 - Built and deployed real-time ML anomaly detection infrastructure processing data streams at scale, leveraging Elastic Stack for observability and Apache Airflow for orchestration. Implemented STIG-compliant security controls and monitoring dashboards, achieving 99.9% uptime.
 - Designed and implemented a neuro-symbolic Foundation model agent for automated hyperparameter tuning that leverages LLMs and VLMs with symbolic reasoning, achieving expert-level performance with 20% efficiency improvement on distributed extended Kalman filter optimization for multi-UAS positioning.
 - Designed a secure, blockchain access control and data integrity framework using containerized ML workflows (Docker/Kubernetes), and NDN architecture for encrypted data exchange.
- 2018 – 2023 ◇ **Research Assistant** SUNY Research Foundation, Binghamton University, Binghamton, NY, USA.
- Designed and deployed a novel deepfake detection framework leveraging Electrical Network Frequency (ENF) signatures extracted from audio/video recordings, optimized for heterogeneous and low-power edge devices.
 - Built and open-sourced [pyenf](#), a Python library for multimodal ENF signal extraction and authentication with modular API design, with support for modular signal processing pipelines.
 - Integrated authentication system into blockchain-enabled smart surveillance platforms, ensuring verifiable data provenance in decentralized public safety infrastructures, and real-time media authentication using ENF as consensus.
 - Developed GAN fingerprinting techniques that leveraged frequency and artifact-based cues to detect AI-generated imagery, contributing to the early identification of synthetic media threats.

Skills

- Deep Learning/LLMs ◇ Pytorch, Tensorflow, FastAI, Hugging Face Transformers, WandB, PEFT, Transformer-lens, NLTK, scikit-learn, Prompt Engineering, FastAPI.
- Coding ◇ Python, C/C++, C#, Shell, Bash, MATLAB, Powershell, SQL, $\LaTeX$, Neo4j and Cypher.

Skills (continued)

Network and Security	◇ DHCP, SSH, VPN, DNS, Port-Forwarding, DMZ N/W, NMAP scan, Hyperledger Fabric, Pentesting using Metasploit framework, OpenDaylight for SDN, LibreNMS, Cacti Server, Tenable Nessus Scanner and DISA STIGs.
Virtualization	◇ ESXi, Vyos, Cisco Nexus, HyperV, Xen, OpenStack, OpenMano, Docker, Kubernetes, Proxmox, GCP and AWS platforms
Misc.	◇ Grafana, Kibana, Elasticsearch, Logstash, Apache Airflow, Wireshark, Git/GitHub.

Education

2017 – 2023	◇ Ph.D, Electrical & Computer Engineering , Binghamton University-SUNY, Binghamton, NY, USA. Dissertation title: <i>Lightweight Multimedia Authentication at the Edge using Environmental Fingerprint</i> Advisor: Dr. Yu Chen
2015 – 2016	◇ MS, Electrical & Computer Engineering , Binghamton University-SUNY, Binghamton, NY, USA. Thesis title: <i>iCrawl: A high interaction client honeypot system.</i> Advisor: Dr. Andrey Dolgikh
2011 – 2015	◇ B.Tech, Electronics & Communications Engineering , SASTRA University, Tamil Nadu, India.

Teaching Experience

Graduate Instructor	◇ Network Computer Security (EECE-480F) ,
Teaching Assistant	◇ Cyber Physical Systems (EECE-480A), Cryptography and Information Security (EECE-405/560), Linear Algebra and Engg Programming (EECE-212) ,

Research Publications

Journal Articles

- 1 Ogunbunmi, S., Chen, Y., Zhao, Q., **Nagothu, D.**, Wei, S., Chen, G., & Blasch, E. (2025). Interest Flooding Attacks in Named Data Networking and Mitigations: Recent Advances and Challenges. *Future Internet*, 17(8), 357. Publisher: Multidisciplinary Digital Publishing Institute. [doi:10.3390/fi17080357](https://doi.org/10.3390/fi17080357)
- 2 Zhao, Q., **Nagothu, D.**, Tian, X., Chen, G., Pham, K. D., & Blasch, E. (2025). Sd-sat: Software-defined multi-constellation satellite communication traffic management framework. *IET Conference Proceedings*, 2024(31), 71–78. Publisher: The Institution of Engineering and Technology. [doi:10.1049/icp.2024.4615](https://doi.org/10.1049/icp.2024.4615)
- 3 Xu, R., **Nagothu, D.**, Chen, Y., Aved, A., Ardiles-Cruz, E., & Blasch, E. (2024). A Secure Interconnected Autonomous System Architecture for Multi-Domain IoT Ecosystems. *IEEE Communications Magazine*, 62(7), 52–57. Conference Name: IEEE Communications Magazine. [doi:10.1109/MCOM.001.2300354](https://doi.org/10.1109/MCOM.001.2300354)
- 4 Qu, Q., Hatami, M., Xu, R., **Nagothu, D.**, Chen, Y., Li, X., ... Chen, G. (2024). The microverse: A task-oriented edge-scale metaverse. *Future Internet*, "16"(2), 60. [doi:10.3390/fi16020060](https://doi.org/10.3390/fi16020060)
- 5 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2022a). Defakepro: Decentralized deepfake attacks detection using enf authentication. *IT Professional*, 24(5), 46–52. [doi:10.1109/MITP.2022.3172653](https://doi.org/10.1109/MITP.2022.3172653)
- 6 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2022b). Deterring deepfake attacks with an electrical network frequency fingerprints approach. *Future Internet*, 14(5), 125. [doi:10.3390/fi14050125](https://doi.org/10.3390/fi14050125)
- 7 Xu, R., **Nagothu, D.**, & Chen, Y. (2021a). Decentralized video input authentication as an edge service for smart cities. *IEEE Consumer Electronics Magazine*, 10(6), 76–82. [doi:10.1109/MCE.2021.3062564](https://doi.org/10.1109/MCE.2021.3062564)
- 8 Xu, R., **Nagothu, D.**, & Chen, Y. (2021b). Econledger: A proof-of-enf consensus based lightweight distributed ledger for iomt networks. *Future Internet*, 13(10), 248. [doi:10.3390/fi13100248](https://doi.org/10.3390/fi13100248)

^oThe complete list of publications is available on my [Google Scholar](#) page

- 9 **Nagothu, D.**, Chen, Y., Aved, A., & Blasch, E. (2021). Authenticating video feeds using electric network frequency estimation at the edge. *EAI Endorsed Transactions on Security and Safety*, "7"(24).
doi:10.4108/eai.4-2-2021.168648
- 10 Xu, R., Nikouei, S. Y., **Nagothu, D.**, Fitwi, A., & Chen, Y. (2020). Blendsps: A blockchain-enabled decentralized smart public safety system. *Smart Cities*, 3(3), 928–951. doi:10.3390/smartcities3030047
- 11 **Nagothu, D.**, Chen, Y., Blasch, E., Aved, A., & Zhu, S. (2019). Detecting malicious false frame injection attacks on surveillance systems at the edge using electrical network frequency signals. *Sensors (Basel)*, 19(11), 1–19.
doi:10.3390/s19112424

Conference Proceedings

- 1 Hatami, M., Qu, Q., **Nagothu, D.**, Mohammadi, J., Chen, Y., Ardiles-Cruz, E., & Blasch, E. (2025). Securing Smart Grid Digital Twins via Real-World ENF Anchors Against Deepfake Attacks. (pp. 392–397).
doi:10.1109/PerComWorkshops65533.2025.00097
- 2 Xu, R., Liu, X., **Nagothu, D.**, Qu, Q., & Chen, Y. (2025). Detecting Manipulated Digital Entities Through Real-World Anchors. In L. Barolli (Ed.), *Advanced Information Networking and Applications* (pp. 450–461).
doi:10.1007/978-3-031-87784-1_41
- 3 Pazytkarim, A., **Nagothu, D.**, & Chen, Y. (2024). A lightweight deep learning model for rapid detection of fabricated ENF signals from audio sources. In *Disruptive Technologies in Information Sciences VIII* (Vol. 13058, pp. 363–375). doi:10.1117/12.3013456
- 4 Poredi, N., Sudarsan, M., Solomon, E., **Nagothu, D.**, & Chen, Y. (2024). Generative adversarial networks-based AI-generated imagery authentication using frequency domain analysis. In *Disruptive Technologies in Information Sciences VIII* (Vol. 13058, pp. 376–390). doi:10.1117/12.3013240
- 5 Zhao, X., **Nagothu, D.**, & Chen, Y. (2024). A homogeneous low-resolution face recognition method using correlation features at the edge. In *Sensors and Systems for Space Applications XVII* (Vol. 13062, pp. 107–123).
doi:10.1117/12.3008368
- 6 Poredi, N., **Nagothu, D.**, & Chen, Y. (2024). Authenticating ai-generated social media images using frequency domain analysis. In *2024 IEEE 21st Consumer Communications & Networking Conference (CCNC)* (pp. 534–539). ISSN: 2331-9860. doi:10.1109/CCNC51664.2024.10454640
- 7 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Ardiles-Cruz, E. (2023). Application of Electrical Network Frequency as an Entropy Generator in Distributed Systems. In *NAECON 2023 - IEEE National Aerospace and Electronics Conference* (pp. 233–238). ISSN: 2379-2027. doi:10.1109/NAECON58068.2023.10365792
- 8 Parker, J., **Nagothu, D.**, & Chen, Y. (2023). Decentralized Vehicular Identification and Tracking on Lightweight IoT Edge Nodes. In *NAECON 2023 - IEEE National Aerospace and Electronics Conference* (pp. 198–203). ISSN: 2379-2027. doi:10.1109/NAECON58068.2023.10365952
- 9 **Nagothu, D.**, Xu, R., & Chen, Y. (2023). DEMA: Decentralized electrical network frequency map for social media authentication. In *Disruptive Technologies in Information Sciences VII* (Vol. 12542, pp. 57–72).
doi:10.1117/12.2663303
- 10 Poredi, N., **Nagothu, D.**, & Chen, Y. (2023). AUSOME: Authenticating social media images using frequency analysis. In *Disruptive Technologies in Information Sciences VII* (Vol. 12542, pp. 44–56). doi:10.1117/12.2663296
- 11 Poredi, N., **Nagothu, D.**, Chen, Y., Li, X., Aved, A., Ardiles-Cruz, E., & Blasch, E. (2022). Robustness of electrical network frequency signals as a fingerprint for digital media authentication. In *2022 IEEE 24th International Workshop on Multimedia Signal Processing (MMSP)* (pp. 1–6). doi:10.1109/MMSP55362.2022.9949315
- 12 **Nagothu, D.**, Dimock, D., Kulesza, A., Yang, H., & Chen, Y. (2022). A distributed crawler for iovt-based public safety surveillance exploring the spatio-temporal correlation. In *Sensors and systems for space applications xv* (Vol. 12121, pp. 18–28). doi:10.1117/12.2618909
- 13 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2021a). Detecting compromised edge smart cameras using lightweight environmental fingerprint consensus. In *Proceedings of the 19th ACM conference on embedded networked sensor systems* (pp. 505–510). doi:10.1145/3485730.3493684

- 14 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2021b). Defake: Decentralized enf-consensus based deepfake detection in video conferencing. In *Ieee 23rd international workshop on multimedia signal processing*.
doi:10.1109/MMSP53017.2021.9733503
- 15 Quan, W., **Nagothu, D.**, Poredi, N., & Chen, Y. (2021). Crip: An efficient critical pixels identification algorithm for fast one-pixel attacks. In *Sensors and systems for space applications xiv* (Vol. 11755, pp. 83–99).
doi:10.1117/12.2581377
- 16 Rosenberg, M., Burns, J. H., **Nagothu, D.**, & Chen, Y. (2020). Enabling continuous operations for uavs with an autonomous service network infrastructure. In *Sensors and systems for space applications xiii* (Vol. 11422, pp. 165–179). doi:10.1117/12.2565866
- 17 Fitwi, A. H., **Nagothu, D.**, Chen, Y., & Blasch, E. (2019). A distributed agent-based framework for a constellation of drones in a military operation. In *Proc. - winter simul. conf.* (Vol. 2019-Decem).
doi:10.1109/WSC40007.2019.9004907
- 18 **Nagothu, D.**, Schwell, J., Chen, Y., Blasch, E., & Zhu, S. (2019). A study on smart online frame forging attacks against video surveillance system. In *Proc. spie - int. soc. opt. eng.* (Vol. 11017). doi:10.1117/12.2519005
- 19 **Nagothu, D.**, Xu, R., Nikouei, S. Y., & Chen, Y. (2019). A microservice-enabled architecture for smart surveillance using blockchain technology. In *2018 ieee int. smart cities conf. isc2 2018*. doi:10.1109/ISC2.2018.8656968
- 20 Nikouei, S. Y., Xu, R., **Nagothu, D.**, Chen, Y., Aved, A., & Blasch, E. (2019). Real-time index authentication for event-oriented surveillance video query using blockchain. In *2018 ieee int. smart cities conf. isc2 2018*.
doi:10.1109/ISC2.2018.8656668
- 21 **Nagothu, D.**, & Dolgikh, A. (2017). Icrawl: A visual high interaction web crawler. In *Lect. notes comput. sci. (including subser. lect. notes artif. intell. lect. notes bioinformatics)* (Vol. 10446 LNCS).
doi:10.1007/978-3-319-65127-9_8

Book Chapters

- 1 Xu, R., **Nagothu, D.**, Chen, Y., Xu, R., **Nagothu, D.**, & Chen, Y. (2024). AR-Edge: Autonomous and Resilient Edge Computing Architecture for Smart Cities. doi:10.5772/intechopen.1005876
- 2 **Nagothu, D.**, Poredi, N., & Chen, Y. (2022). Evolution of attacks on intelligent surveillance systems and effective detection techniques. doi:10.5772/intechopen.105958
- 3 Xu, R., **Nagothu, D.**, & Chen, Y. (2022). Ecom: Epoch randomness-based consensus committee configuration for iot blockchains. In K. Daimi, I. Dionysiou, & N. El Madhoun (Eds.), *Principles and practice of blockchains* (pp. 135–154). doi:10.1007/978-3-031-10507-4_7
- 4 **Nagothu, D.**, Xu, R., Nikouei, S. Y., Zhao, X., & Chen, Y. (2020). Smart surveillance for public safety enabled by edge computing. (pp. 409–433). doi:10.1049/PBPC033E_ch19

Books

- 1 **Nagothu, D.**, & Chen, Y. (2023). *Authentication of video feeds in smart edge surveillance networks* (C. Olson, Ed.). Bellingham, Washington 98227-0010: SPIE Press.

Dissertation and Thesis

- 1 **Nagothu, D.** (2023). *Lightweight Multimedia Authentication at the Edge Using Environmental Fingerprint* (Ph.D. State University of New York at Binghamton, United States – New York). ISBN: 9798380566988. Retrieved October 28, 2023, from <https://www.proquest.com/docview/2872097834/abstract/A49AAD7CD800446BPQ/1>
- 2 **Nagothu, D.** (2016). *Icrawl: A high interaction client honeypot system* (M.S. State University of New York at Binghamton, United States – New York).

Professional Activities

Reviewer for Journals

- ◇ IEEE Transactions on Pattern Analysis and Machine Intelligence
- ◇ IEEE Transactions on Multimedia Computing Communications
- ◇ IEEE Transactions on Dependable and Services Computing
- ◇ IEEE Transactions on Services Computing
- ◇ IEEE Internet of Things Journal
- ◇ SPIE Journal of Electronic Imaging (JEI).
- ◇ Elsevier Computers and Security
- ◇ Expert Systems with Applications
- ◇ IEEE Access
- ◇ IEEE Transactions on Aerospace and Electronic Systems
- ◇ IEEE Transactions on Cloud Computing
- ◇ IEEE Computers
- ◇ Applied Sciences
- ◇ MDPI Sensors

Reviewer for Conferences

- ◇ IEEE International Conference on Computer Communications (INFOCOM).
- ◇ IEEE Communications Magazine (COMMAG)
- ◇ IEEE Global Communications Conference (GLOBECOM) IoT and Sensor Networks (IoTSN).
- ◇ IEEE Global Communications Conference (GLOBECOM) Communication and Information Systems Security (CISS).
- ◇ IEEE Global Communications Conference (GLOBECOM) Communications Software, Services and Multimedia Apps (CSSMA).
- ◇ IEEE International Conference on Wireless and Mobile Computing, Networking And Communications (WiMob).
- ◇ ACM International Workshop on Blockchain-enabled Networked Sensor Systems (BlockSys)
- ◇ IEEE International Smart Cities Conference (ISC2).
- ◇ IEEE International Conference on Cloud Networking (CloudNet)
- ◇ IEEE International Conference on Communications (ICC)
- ◇ Knowledge based Systems (KNOSYS)
- ◇ Scientific Reports

Miscellaneous Experience

Awards and Achievements

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| 2025 | <ul style="list-style-type: none">◇ Best Paper Award, Best Workshop Paper at IEEE Percom SPT-IoT Conference. (Securing Smart Grid Digital Twins via Real-World ENF Anchors against DeepFake Attacks)◇ Finalist, xTech AI Competition, for the project "GRADIENT", focused on synthetic data-based feature engineering for robust AI model performance.◇ Winner IEEE AESS Cybersecurity Challenge: Authored the winning challenge proposal "Federated AI for Resilient Avionics and Drone Operations," establishing a federated learning benchmark for resilient aerospace and drone cybersecurity research. |
| 2020 | <ul style="list-style-type: none">◇ GSEA, Graduate Student Award for Excellence in Teaching (Courses - Network Computer Security and Cyber Physical Systems). |